

offline_evaluation

June 18, 2025

0.1 Offline Evaluation

in this notebook, I aim to work on following ideas:

- ☒ run experiments with different parameters
- ☐ list experiment artifacts, metrics and results in general
- ☒ run A/B testing pipeline in the notebook by passing two different models loaded from the model registry, and splitted test data
- ☒ run all models in an experiment against a part of data (here one of A_B_1 or A_B_2)

```
[51]: dataset = "data/  
    ↪data_train_features_need_preprocessing_salary_less_than_500k_and_above_1k.  
    ↪parquet"  
dataset_name = "linkedin_joblist_salary_less_than_500k_and_above_1k"  
  
flow = "pipelines/experiment_1_decision_tree_fault_tolerance.py"
```

0.1.1 Split dataset

```
[52]: import pandas as pd  
from sklearn.model_selection import train_test_split  
  
seed = 42  
  
df = pd.read_parquet(dataset)  
  
train_df, temp_df = train_test_split(df, test_size=0.2, random_state=seed)  
A_B_1, A_B_2 = train_test_split(temp_df, test_size=0.5, random_state=seed)  
  
train_df.to_parquet('data/train.parquet', index=False)  
A_B_1.to_parquet('data/A_B_1.parquet', index=False)  
A_B_2.to_parquet('data/A_B_2.parquet', index=False)
```

```
[48]: dataset_train = "data/train.parquet"  
dataset_A_B_1 = "data/A_B_1.parquet"  
dataset_A_B_2 = "data/A_B_2.parquet"
```

0.1.2 Run experiments with different parameters

```
[45]: from metaflow import Runner
```

```
[46]: # it is set by default
# max_depth = 30
run_id="A_B_test_control"
model_name="A_B_control_model"
with Runner(flow).run(dataset='data/train.parquet', dataset_name=dataset_name,
    model_name=model_name) as running:
    print(f'{running.run} finished')
```

Metaflow 2.15.9 executing DTRFlow for user:mahdikhashan

Validating your flow...

The graph looks good!

Running pylint...

Pylint not found, so extra checks are disabled.

Including file data/train.parquet of size 50MB

Attempting to use MLflow Tracking URI: None

2025-06-18 18:10:08.336 Workflow starting (run-id 1750263006872182):

2025-06-18 18:10:08.350 [1750263006872182/start/1 (pid 25606)] Task is starting.

2025-06-18 18:10:08.760 [1750263006872182/start/1 (pid 25606)] Attempting to use

MLflow Tracking URI: None

2025-06-18 18:10:08.829 [1750263006872182/start/1 (pid 25606)] Starting flow
'DTRFlow' with run ID '1750263006872182'

2025-06-18 18:10:08.893 [1750263006872182/start/1 (pid 25606)] Task finished
successfully.

2025-06-18 18:10:08.899 [1750263006872182/load_dataset/2 (pid 25613)] Task is
starting.

2025-06-18 18:10:09.277 [1750263006872182/load_dataset/2 (pid 25613)] Attempting
to use MLflow Tracking URI: None

2025-06-18 18:10:09.531 [1750263006872182/load_dataset/2 (pid 25613)] Loading
dataset from IncludeFile: 52966323

2025-06-18 18:10:09.709 [1750263006872182/load_dataset/2 (pid 25613)] Dataset
loaded successfully. Shape: (23101, 10)

2025-06-18 18:10:09.709 [1750263006872182/load_dataset/2 (pid 25613)] First 5
rows sample:

2025-06-18 18:10:09.715 [1750263006872182/load_dataset/2 (pid 25613)]

company_name ... normalized_salary

2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] 0

Strategic Staffing Solutions ... 68640.0

2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] 1

Cognizant ... 154450.0

2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] 2

Allen & Overy ... 110000.0

2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] 3

ISITE TECHNOLOGIES ... 88400.0

2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] 4 Siemens

```

EDA (Siemens Digital Industries Software) ... 203550.0
2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)]
2025-06-18 18:10:09.905 [1750263006872182/load_dataset/2 (pid 25613)] [5 rows x
10 columns]
2025-06-18 18:10:09.906 [1750263006872182/load_dataset/2 (pid 25613)] Task
finished successfully.
2025-06-18 18:10:09.910 [1750263006872182/dataset_is_large_enough/3 (pid 25620)]
Task is starting.
2025-06-18 18:10:10.286 [1750263006872182/dataset_is_large_enough/3 (pid 25620)]
Attempting to use MLflow Tracking URI: None
2025-06-18 18:10:10.796 [1750263006872182/dataset_is_large_enough/3 (pid 25620)]
Task finished successfully.
2025-06-18 18:10:10.802 [1750263006872182/validate_dataset/4 (pid 25630)] Task
is starting.
2025-06-18 18:10:11.210 [1750263006872182/validate_dataset/4 (pid 25630)]
Attempting to use MLflow Tracking URI: None
2025-06-18 18:10:11.277 [1750263006872182/validate_dataset/4 (pid 25630)]
Attempting to add project root to sys.path: /Users/mahdikhashan/tmp/tu-mlops
2025-06-18 18:10:11.278 [1750263006872182/validate_dataset/4 (pid 25630)]
'/Users/mahdikhashan/tmp/tu-mlops' added to sys.path.
2025-06-18 18:10:11.278 [1750263006872182/validate_dataset/4 (pid 25630)]
*****
2025-06-18 18:10:11.278 [1750263006872182/validate_dataset/4 (pid 25630)]
Attempting validation setup...
2025-06-18 18:10:11.278 [1750263006872182/validate_dataset/4 (pid 25630)]
*****
2025-06-18 18:10:11.278 [1750263006872182/validate_dataset/4 (pid 25630)]
Successfully imported test fixture setup from tests.conftest.
2025-06-18 18:10:11.568 [1750263006872182/validate_dataset/4 (pid 25630)]
Passing DataFrame to fixture setup mechanism...
2025-06-18 18:10:11.568 [1750263006872182/validate_dataset/4 (pid 25630)]
*****
2025-06-18 18:10:11.569 [1750263006872182/validate_dataset/4 (pid 25630)] Debug
log - First row's title:
2025-06-18 18:10:11.569 [1750263006872182/validate_dataset/4 (pid 25630)]
Brokerage Escalation Representatives (Series 7 Required)
2025-06-18 18:10:11.629 [1750263006872182/validate_dataset/4 (pid 25630)]
*****
2025-06-18 18:10:11.629 [1750263006872182/validate_dataset/4 (pid 25630)]
Running pytest on directory: /Users/mahdikhashan/tmp/tu-mlops/tests
2025-06-18 18:10:11.629 [1750263006872182/validate_dataset/4 (pid 25630)]
===== test session starts =====
2025-06-18 18:10:11.630 [1750263006872182/validate_dataset/4 (pid 25630)]
platform darwin -- Python 3.11.10, pytest-8.3.2, pluggy-1.6.0 --
/Users/mahdikhashan/tmp/tu-mlops/.venv/bin/python
2025-06-18 18:10:11.630 [1750263006872182/validate_dataset/4 (pid 25630)]
cachedir: .pytest_cache
2025-06-18 18:10:13.285 [1750263006872182/validate_dataset/4 (pid 25630)]

```

```

rootdir: /Users/mahdikhashan/tmp/tu-mlops
2025-06-18 18:10:13.286 [1750263006872182/validate_dataset/4 (pid 25630)]
configfile: pyproject.toml
2025-06-18 18:10:13.286 [1750263006872182/validate_dataset/4 (pid 25630)]
plugins: anyio-4.9.0
2025-06-18 18:10:13.286 [1750263006872182/validate_dataset/4 (pid 25630)]
collecting ... 1.3.11
2025-06-18 18:10:13.293 [1750263006872182/validate_dataset/4 (pid 25630)]
collected 5 items
2025-06-18 18:10:13.295 [1750263006872182/validate_dataset/4 (pid 25630)]
2025-06-18 18:10:13.805 [1750263006872182/validate_dataset/4 (pid 25630)]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 2949.58it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 2548.18it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 3421.13it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 3226.39it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 2155.07it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 2106.37it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2977.32it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2931.54it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2711.91it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2675.79it/s]
2025-06-18 18:10:13.811 [1750263006872182/validate_dataset/4 (pid 25630)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_title PASSED
2025-06-18 18:10:14.190 [1750263006872182/validate_dataset/4 (pid 25630)]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 15621.24it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 10578.32it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 13691.96it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 11672.46it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 2783.58it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 2705.30it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 5387.41it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 5312.35it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 5265.16it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 5189.49it/s]
2025-06-18 18:10:14.195 [1750263006872182/validate_dataset/4 (pid 25630)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_description
PASSED
2025-06-18 18:10:14.550 [1750263006872182/validate_dataset/4 (pid 25630)]
tests/test_validate_dataset.py::test_validate_max_salary_in_range PASSED
2025-06-18 18:10:14.889 [1750263006872182/validate_dataset/4 (pid 25630)]
tests/test_validate_dataset.py::test_validate_min_salary_in_range PASSED
2025-06-18 18:10:14.889 [1750263006872182/validate_dataset/4 (pid 25630)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]

```

```

Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 16545.58it/s]
Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 12652.50it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 16320.25it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 14580.43it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 10965.50it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 10286.46it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 9834.24it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 9198.04it/s]
2025-06-18 18:10:15.215 [1750263006872182/validate_dataset/4 (pid 25630)]
Calculating Metrics: 0%|           | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 0%|           | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 40%|          | 2/5 [00:00<00:00, 19831.22it/s]
Calculating Metrics: 40%|          | 2/5 [00:00<00:00, 14588.88it/s]
Calculating Metrics: 60%|          | 3/5 [00:00<00:00, 18315.74it/s]
Calculating Metrics: 60%|          | 3/5 [00:00<00:00, 16215.09it/s]
Calculating Metrics: 100%|         | 5/5 [00:00<00:00, 16033.27it/s]
Calculating Metrics: 100%|         | 5/5 [00:00<00:00, 14883.97it/s]
Calculating Metrics: 100%|         | 5/5 [00:00<00:00, 14169.95it/s]
Calculating Metrics: 100%|         | 5/5 [00:00<00:00, 13123.60it/s]
2025-06-18 18:10:15.217 [1750263006872182/validate_dataset/4 (pid 25630)]
Calculating Metrics: 0%|           | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|           | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 17734.90it/s]
Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 13421.77it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 16958.10it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 15087.42it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 1461.56it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 1441.96it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 1430.65it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 1414.96it/s]
2025-06-18 18:10:15.220 [1750263006872182/validate_dataset/4 (pid 25630)]
tests/test_validate_dataset.py::test_validate_column_values_in_list PASSED
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
===== warnings summary =====
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095:
ChangedInMarshmallow4Warning: `Number` field should not be instantiated. Use
`Integer`, `Float`, or `Decimal` instead.
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
config_version: fields.Number = fields.Number(
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)] --
Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html

```

```

2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)]
===== 5 passed, 1 warning in 3.59s =====
2025-06-18 18:10:15.221 [1750263006872182/validate_dataset/4 (pid 25630)] Pytest
validation successful!
2025-06-18 18:10:15.876 [1750263006872182/validate_dataset/4 (pid 25630)] Task
finished successfully.
2025-06-18 18:10:15.887 [1750263006872182/preprocess_training_data/5 (pid
25644)] Task is starting.
2025-06-18 18:10:16.424 [1750263006872182/preprocess_training_data/5 (pid
25644)] Attempting to use MLflow Tracking URI: None
2025-06-18 18:10:17.073 [1750263006872182/preprocess_training_data/5 (pid
25644)] Fitting preprocessor on training data...
2025-06-18 18:10:21.648 [1750263006872182/preprocess_training_data/5 (pid
25644)] Preprocessor fitted.
2025-06-18 18:10:26.042 [1750263006872182/preprocess_training_data/5 (pid
25644)] Transforming training data...
2025-06-18 18:10:26.043 [1750263006872182/preprocess_training_data/5 (pid
25644)] Training data transformed. Shape: (18480, 12895)
2025-06-18 18:10:26.051 [1750263006872182/preprocess_training_data/5 (pid
25644)] Successfully retrieved 12895 feature names.
2025-06-18 18:10:26.051 [1750263006872182/preprocess_training_data/5 (pid
25644)] Sample feature names: ['title_tfidf__00' 'title_tfidf__000'
'title_tfidf__10' 'title_tfidf__100'
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] 'title_tfidf__100k' 'title_tfidf__11' 'title_tfidf__12'
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] 'title_tfidf__1st' 'title_tfidf__20' 'title_tfidf__2024'
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] 'title_tfidf__2025' 'title_tfidf__24' 'title_tfidf__25'
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] 'title_tfidf__2nd' 'title_tfidf__30' 'title_tfidf__30pm'
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] 'title_tfidf__3rd' 'title_tfidf__40' 'title_tfidf__50'
'title_tfidf__500']
2025-06-18 18:10:26.052 [1750263006872182/preprocess_training_data/5 (pid
25644)] Processed Training DataFrame created.
2025-06-18 18:10:26.077 [1750263006872182/preprocess_training_data/5 (pid
25644)] title_tfidf__00 ... cat_onehot__location_Zionsville, IN
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] 15628          0.0 ...          0.0
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] 4015          0.0 ...          0.0
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] 4453          0.0 ...          0.0
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] 8845          0.0 ...          0.0
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] 9195          0.0 ...          0.0

```

```

2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)]
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] [5 rows x 12895 columns]
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)]
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] Transforming test data...
2025-06-18 18:10:27.049 [1750263006872182/preprocess_training_data/5 (pid
25644)] Test data transformed. Shape: (4621, 12895)
2025-06-18 18:10:27.068 [1750263006872182/preprocess_training_data/5 (pid
25644)] Processed Test DataFrame created.
2025-06-18 18:10:27.068 [1750263006872182/preprocess_training_data/5 (pid
25644)] title_tfidf__00 ... cat_onehot__location_Zionsville, IN
2025-06-18 18:10:30.732 [1750263006872182/preprocess_training_data/5 (pid
25644)] 22136          0.0 ...          0.0
2025-06-18 18:10:30.732 [1750263006872182/preprocess_training_data/5 (pid
25644)] 15278          0.0 ...          0.0
2025-06-18 18:10:30.733 [1750263006872182/preprocess_training_data/5 (pid
25644)] 22573          0.0 ...          0.0
2025-06-18 18:10:30.733 [1750263006872182/preprocess_training_data/5 (pid
25644)] 8310          0.0 ...          0.0
2025-06-18 18:10:30.733 [1750263006872182/preprocess_training_data/5 (pid
25644)] 22669          0.0 ...          0.0
2025-06-18 18:10:30.733 [1750263006872182/preprocess_training_data/5 (pid
25644)]
2025-06-18 18:10:30.733 [1750263006872182/preprocess_training_data/5 (pid
25644)] [5 rows x 12895 columns]
2025-06-18 18:10:30.734 [1750263006872182/preprocess_training_data/5 (pid
25644)] Task finished successfully.
2025-06-18 18:10:30.741 [1750263006872182/train_model/6 (pid 25658)] Task is
starting.
2025-06-18 18:10:31.306 [1750263006872182/train_model/6 (pid 25658)] Attempting
to use MLflow Tracking URI: None
2025-06-18 18:10:32.660 [1750263006872182/train_model/6 (pid 25658)] in mlflow
context
2025-06-18 18:10:32.695 [1750263006872182/train_model/6 (pid 25658)] MLflow Run
ID: 2b94defc163c46e582550f897fd3c01b
2025-06-18 18:10:32.695 [1750263006872182/train_model/6 (pid 25658)] MLflow Run
Name: 1750263006872182
2025-06-18 18:10:32.695 [1750263006872182/train_model/6 (pid 25658)] MLflow
Experiment Dataset Name: linkedin_joblist_salary_less_than_500k_and_above_1k
2025-06-18 18:10:32.695 [1750263006872182/train_model/6 (pid 25658)] Training
Decision Tree Regressor...
2025-06-18 18:10:32.699 [1750263006872182/train_model/6 (pid 25658)] <flow
DTRFlow step train_model> failed:
2025-06-18 18:10:32.711 [1750263006872182/train_model/6 (pid 25658)] Internal
error

```

```

2025-06-18 18:10:32.714 [1750263006872182/train_model/6 (pid 25658)] Traceback
(most recent call last):
2025-06-18 18:10:32.714 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli.py", line 619, in main
2025-06-18 18:10:32.714 [1750263006872182/train_model/6 (pid 25658)]
start(auto_envvar_prefix="METAFLOW", obj=state)
2025-06-18 18:10:32.714 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 829, in __call__
2025-06-18 18:10:32.714 [1750263006872182/train_model/6 (pid 25658)] return
self.main(args, kwargs)
2025-06-18 18:10:32.997 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 782, in main
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] rv =
self.invoke(ctx)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli_components/utils.py", line 69, in invoke
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] return
_process_result(sub_ctx.command.invoke(sub_ctx))
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 1066, in invoke
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] return
ctx.invoke(self.callback, ctx.params)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 610, in invoke
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] return
callback(args, kwargs)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/tracing/__init__.py", line 27, in wrapper_func
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] return
func(args, kwargs)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]

```



```

~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/decorators.py", line 21, in new_func
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] return
f(get_current_context(), args, kwargs)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
~~~~~
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli_components/step_cmd.py", line 167, in step
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
task.run_step(
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 672, in run_step
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
self._exec_step_function(step_func)
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 62, in _exec_step_function
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
step_function()
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] File
"/Users/mahdikhashan/tmp/tu-
mlops/pipelines/experiment_1_decision_tree_fault_tolerance.py", line 363, in
train_model
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)] raise
Fault("model fitting failed - it should be handled, why!!!")
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
utils.FaultException.Fault: model fitting failed - it should be handled, why!!!
2025-06-18 18:10:32.998 [1750263006872182/train_model/6 (pid 25658)]
2025-06-18 18:10:32.999 [1750263006872182/train_model/6 (pid 25658)] Task
failed.
2025-06-18 18:10:33.001 [1750263006872182/train_model/6 (pid 25671)] Task is
starting (retry).
2025-06-18 18:10:33.628 [1750263006872182/train_model/6 (pid 25671)] Attempting
to use MLflow Tracking URI: None
2025-06-18 18:10:34.462 [1750263006872182/train_model/6 (pid 25671)] in mlflow
context
2025-06-18 18:10:34.493 [1750263006872182/train_model/6 (pid 25671)] MLflow Run
ID: e76424485cf249f7bcf640b1b936cb15
2025-06-18 18:10:34.493 [1750263006872182/train_model/6 (pid 25671)] MLflow Run
Name: 1750263006872182
2025-06-18 18:10:34.494 [1750263006872182/train_model/6 (pid 25671)] MLflow
Experiment Dataset Name: linkedin_joblist_salary_less_than_500k_and_above_1k
2025-06-18 18:10:34.494 [1750263006872182/train_model/6 (pid 25671)] Training
Decision Tree Regressor...

```

2025-06-18 18:10:50.481 [1750263006872182/train_model/6 (pid 25671)] Model training complete.

2025-06-18 18:10:50.959 [1750263006872182/train_model/6 (pid 25671)] Making predictions on the test set...

2025-06-18 18:10:50.959 [1750263006872182/train_model/6 (pid 25671)]

2025-06-18 18:10:50.965 [1750263006872182/train_model/6 (pid 25671)] Evaluating model performance on the test set:

2025-06-18 18:10:50.965 [1750263006872182/train_model/6 (pid 25671)] Mean Absolute Error (MAE): 32557.17

2025-06-18 18:10:52.118 [1750263006872182/train_model/6 (pid 25671)] Root Mean Squared Error (RMSE): 47765.77

2025-06-18 18:10:52.118 [1750263006872182/train_model/6 (pid 25671)] R-squared (R^2): 0.3192

2025-06-18 18:10:52.118 [1750263006872182/train_model/6 (pid 25671)]

2025-06-18 18:10:52.132 [1750263006872182/train_model/6 (pid 25671)] Root Mean Squared Error (RMSE) on Training Set: 37572.44

2025-06-18 18:10:52.132 [1750263006872182/train_model/6 (pid 25671)] logging current model version.

2025-06-18 18:12:34.302 [1750263006872182/train_model/6 (pid 25671)] 2025/06/18 18:12:34 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead.

2025-06-18 18:12:37.373 [1750263006872182/train_model/6 (pid 25671)] Successfully registered model 'A_B_control_model'.

2025-06-18 18:12:37.376 [1750263006872182/train_model/6 (pid 25671)] Created version '1' of model 'A_B_control_model'.

2025-06-18 18:12:37.381 [1750263006872182/train_model/6 (pid 25671)] Current model (R^2 : 0.3192) logged as A_B_control_model.

2025-06-18 18:12:41.324 [1750263006872182/train_model/6 (pid 25671)] Task finished successfully.

2025-06-18 18:12:41.332 [1750263006872182/evaluate_robustness/7 (pid 25760)] Task is starting.

2025-06-18 18:12:41.832 [1750263006872182/evaluate_robustness/7 (pid 25760)] Attempting to use MLflow Tracking URI: None

2025-06-18 18:12:42.211 [1750263006872182/evaluate_robustness/7 (pid 25760)] R2 metric found, comparing with the baseline

2025-06-18 18:12:42.212 [1750263006872182/evaluate_robustness/7 (pid 25760)] Current model (R^2 : 0.3192) should be tagged as champion due to higher r2 value than previous one.

2025-06-18 18:12:42.330 [1750263006872182/evaluate_robustness/7 (pid 25760)] Task finished successfully.

2025-06-18 18:12:42.338 [1750263006872182/end/8 (pid 25767)] Task is starting.

2025-06-18 18:12:42.733 [1750263006872182/end/8 (pid 25767)] Attempting to use MLflow Tracking URI: None

2025-06-18 18:12:42.799 [1750263006872182/end/8 (pid 25767)] Flow 'DTRFlow' completed successfully.

2025-06-18 18:12:42.862 [1750263006872182/end/8 (pid 25767)] Task finished successfully.

2025-06-18 18:12:42.863 Done!

Run('DTRFlow/1750263006872182') finished

```
[47]: # second model training with a different parameter
max_depth=5
model_name="A_B_second_model"
with Runner(flow).run(dataset='data/train.parquet', dataset_name=dataset_name,
    ↳max_depth=max_depth, model_name=model_name) as running:
    print(f'{running.run} finished')
```

Metaflow 2.15.9 executing DTRFlow for user:mahdikhashan

Validating your flow...

The graph looks good!

Running pylint...

Pylint not found, so extra checks are disabled.

Including file data/train.parquet of size 50MB

Attempting to use MLflow Tracking URI: None

2025-06-18 18:13:20.178 Workflow starting (run-id 1750263198685669):

2025-06-18 18:13:20.190 [1750263198685669/start/1 (pid 25799)] Task is starting.

2025-06-18 18:13:20.581 [1750263198685669/start/1 (pid 25799)] Attempting to use MLflow Tracking URI: None

2025-06-18 18:13:20.648 [1750263198685669/start/1 (pid 25799)] Starting flow 'DTRFlow' with run ID '1750263198685669'

2025-06-18 18:13:20.715 [1750263198685669/start/1 (pid 25799)] Task finished successfully.

2025-06-18 18:13:20.720 [1750263198685669/load_dataset/2 (pid 25806)] Task is starting.

2025-06-18 18:13:21.193 [1750263198685669/load_dataset/2 (pid 25806)] Attempting to use MLflow Tracking URI: None

2025-06-18 18:13:21.468 [1750263198685669/load_dataset/2 (pid 25806)] Loading dataset from IncludeFile: 52966323

2025-06-18 18:13:21.673 [1750263198685669/load_dataset/2 (pid 25806)] Dataset loaded successfully. Shape: (23101, 10)

2025-06-18 18:13:21.674 [1750263198685669/load_dataset/2 (pid 25806)] First 5 rows sample:

2025-06-18 18:13:21.682 [1750263198685669/load_dataset/2 (pid 25806)] company_name ... normalized_salary

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] 0 Strategic Staffing Solutions ... 68640.0

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] 1 Cognizant ... 154450.0

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] 2 Allen & Overy ... 110000.0

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] 3 ISITE TECHNOLOGIES ... 88400.0

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] 4 Siemens EDA (Siemens Digital Industries Software) ... 203550.0

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)]

2025-06-18 18:13:21.878 [1750263198685669/load_dataset/2 (pid 25806)] [5 rows x

```

10 columns]
2025-06-18 18:13:21.879 [1750263198685669/load_dataset/2 (pid 25806)] Task
finished successfully.
2025-06-18 18:13:21.884 [1750263198685669/dataset_is_large_enough/3 (pid 25813)]
Task is starting.
2025-06-18 18:13:22.308 [1750263198685669/dataset_is_large_enough/3 (pid 25813)]
Attempting to use MLflow Tracking URI: None
2025-06-18 18:13:22.778 [1750263198685669/dataset_is_large_enough/3 (pid 25813)]
Task finished successfully.
2025-06-18 18:13:22.783 [1750263198685669/validate_dataset/4 (pid 25820)] Task
is starting.
2025-06-18 18:13:23.164 [1750263198685669/validate_dataset/4 (pid 25820)]
Attempting to use MLflow Tracking URI: None
2025-06-18 18:13:23.233 [1750263198685669/validate_dataset/4 (pid 25820)]
Attempting to add project root to sys.path: /Users/mahdikhashan/tmp/tu-mlops
2025-06-18 18:13:23.236 [1750263198685669/validate_dataset/4 (pid 25820)]
'/Users/mahdikhashan/tmp/tu-mlops' added to sys.path.
2025-06-18 18:13:23.236 [1750263198685669/validate_dataset/4 (pid 25820)]
*****
2025-06-18 18:13:23.236 [1750263198685669/validate_dataset/4 (pid 25820)]
Attempting validation setup...
2025-06-18 18:13:23.236 [1750263198685669/validate_dataset/4 (pid 25820)]
*****
2025-06-18 18:13:23.236 [1750263198685669/validate_dataset/4 (pid 25820)]
Successfully imported test fixture setup from tests.conftest.
2025-06-18 18:13:23.448 [1750263198685669/validate_dataset/4 (pid 25820)]
Passing DataFrame to fixture setup mechanism...
2025-06-18 18:13:23.448 [1750263198685669/validate_dataset/4 (pid 25820)]
*****
2025-06-18 18:13:23.448 [1750263198685669/validate_dataset/4 (pid 25820)] Debug
log - First row's title:
2025-06-18 18:13:23.448 [1750263198685669/validate_dataset/4 (pid 25820)]
Brokerage Escalation Representatives (Series 7 Required)
2025-06-18 18:13:23.509 [1750263198685669/validate_dataset/4 (pid 25820)]
*****
2025-06-18 18:13:23.509 [1750263198685669/validate_dataset/4 (pid 25820)]
Running pytest on directory: /Users/mahdikhashan/tmp/tu-mlops/tests
2025-06-18 18:13:23.509 [1750263198685669/validate_dataset/4 (pid 25820)]
===== test session starts =====
2025-06-18 18:13:23.510 [1750263198685669/validate_dataset/4 (pid 25820)]
platform darwin -- Python 3.11.10, pytest-8.3.2, pluggy-1.6.0 --
/Users/mahdikhashan/tmp/tu-mlops/.venv/bin/python
2025-06-18 18:13:23.510 [1750263198685669/validate_dataset/4 (pid 25820)]
cachedir: .pytest_cache
2025-06-18 18:13:23.510 [1750263198685669/validate_dataset/4 (pid 25820)]
rootdir: /Users/mahdikhashan/tmp/tu-mlops
2025-06-18 18:13:23.510 [1750263198685669/validate_dataset/4 (pid 25820)]
configfile: pyproject.toml

```

```

2025-06-18 18:13:23.510 [1750263198685669/validate_dataset/4 (pid 25820)]
plugins: anyio-4.9.0
2025-06-18 18:13:24.767 [1750263198685669/validate_dataset/4 (pid 25820)]
collecting ... 1.3.11
2025-06-18 18:13:24.775 [1750263198685669/validate_dataset/4 (pid 25820)]
collected 5 items
2025-06-18 18:13:24.776 [1750263198685669/validate_dataset/4 (pid 25820)]
2025-06-18 18:13:25.264 [1750263198685669/validate_dataset/4 (pid 25820)]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 2614.90it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 2302.03it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 3282.78it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 3161.54it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 2232.20it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 2184.82it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 3048.74it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2992.99it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2964.17it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2927.19it/s]
2025-06-18 18:13:25.269 [1750263198685669/validate_dataset/4 (pid 25820)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_title PASSED
2025-06-18 18:13:25.661 [1750263198685669/validate_dataset/4 (pid 25820)]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 13421.77it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 4473.92it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 6026.30it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 5535.82it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 2348.70it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 2287.47it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 4479.62it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 4421.86it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 4389.11it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 4334.68it/s]
2025-06-18 18:13:25.665 [1750263198685669/validate_dataset/4 (pid 25820)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_description
PASSED
2025-06-18 18:13:26.016 [1750263198685669/validate_dataset/4 (pid 25820)]
tests/test_validate_dataset.py::test_validate_max_salary_in_range PASSED
2025-06-18 18:13:26.016 [1750263198685669/validate_dataset/4 (pid 25820)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 16513.01it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 12576.62it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 16215.09it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 14699.66it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 11554.56it/s]

```

```

Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 10837.99it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 10388.37it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 9703.42it/s]
2025-06-18 18:13:26.392 [1750263198685669/validate_dataset/4 (pid 25820)]
Calculating Metrics: 0%|        | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 0%|        | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 40%|       | 2/5 [00:00<00:00, 11748.75it/s]
Calculating Metrics: 40%|       | 2/5 [00:00<00:00, 9467.95it/s]
Calculating Metrics: 60%|       | 3/5 [00:00<00:00, 12495.44it/s]
Calculating Metrics: 60%|       | 3/5 [00:00<00:00, 11356.42it/s]
Calculating Metrics: 100%|      | 5/5 [00:00<00:00, 11459.85it/s]
Calculating Metrics: 100%|      | 5/5 [00:00<00:00, 10771.20it/s]
Calculating Metrics: 100%|      | 5/5 [00:00<00:00, 10325.71it/s]
Calculating Metrics: 100%|      | 5/5 [00:00<00:00, 9704.54it/s]
2025-06-18 18:13:26.395 [1750263198685669/validate_dataset/4 (pid 25820)]
tests/test_validate_dataset.py::test_validate_min_salary_in_range PASSED
2025-06-18 18:13:26.748 [1750263198685669/validate_dataset/4 (pid 25820)]
Calculating Metrics: 0%|        | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|        | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|       | 2/4 [00:00<00:00, 17084.74it/s]
Calculating Metrics: 50%|       | 2/4 [00:00<00:00, 13005.59it/s]
Calculating Metrics: 75%|       | 3/4 [00:00<00:00, 16469.78it/s]
Calculating Metrics: 75%|       | 3/4 [00:00<00:00, 14699.66it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 1661.93it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 1632.02it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 1615.52it/s]
Calculating Metrics: 100%|      | 4/4 [00:00<00:00, 1595.55it/s]
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
tests/test_validate_dataset.py::test_validate_column_values_in_list PASSED
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
===== warnings summary =====
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095:
ChangedInMarshmallow4Warning: `Number` field should not be instantiated. Use
`Integer`, `Float`, or `Decimal` instead.
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
config_version: fields.Number = fields.Number(
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)] --
Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)]
===== 5 passed, 1 warning in 3.24s =====
2025-06-18 18:13:26.751 [1750263198685669/validate_dataset/4 (pid 25820)] Pytest

```

```

validation successful!
2025-06-18 18:13:27.372 [1750263198685669/validate_dataset/4 (pid 25820)] Task
finished successfully.
2025-06-18 18:13:27.378 [1750263198685669/preprocess_training_data/5 (pid
25836)] Task is starting.
2025-06-18 18:13:27.920 [1750263198685669/preprocess_training_data/5 (pid
25836)] Attempting to use MLflow Tracking URI: None
2025-06-18 18:13:28.628 [1750263198685669/preprocess_training_data/5 (pid
25836)] Fitting preprocessor on training data...
2025-06-18 18:13:33.472 [1750263198685669/preprocess_training_data/5 (pid
25836)] Preprocessor fitted.
2025-06-18 18:13:40.163 [1750263198685669/preprocess_training_data/5 (pid
25836)] Transforming training data...
2025-06-18 18:13:40.165 [1750263198685669/preprocess_training_data/5 (pid
25836)] Training data transformed. Shape: (18480, 12895)
2025-06-18 18:13:40.174 [1750263198685669/preprocess_training_data/5 (pid
25836)] Successfully retrieved 12895 feature names.
2025-06-18 18:13:40.176 [1750263198685669/preprocess_training_data/5 (pid
25836)] Sample feature names: ['title_tfidf__00' 'title_tfidf__000'
'title_tfidf__10' 'title_tfidf__100'
2025-06-18 18:13:40.176 [1750263198685669/preprocess_training_data/5 (pid
25836)] 'title_tfidf__100k' 'title_tfidf__11' 'title_tfidf__12'
2025-06-18 18:13:40.177 [1750263198685669/preprocess_training_data/5 (pid
25836)] 'title_tfidf__1st' 'title_tfidf__20' 'title_tfidf__2024'
2025-06-18 18:13:40.177 [1750263198685669/preprocess_training_data/5 (pid
25836)] 'title_tfidf__2025' 'title_tfidf__24' 'title_tfidf__25'
2025-06-18 18:13:40.177 [1750263198685669/preprocess_training_data/5 (pid
25836)] 'title_tfidf__2nd' 'title_tfidf__30' 'title_tfidf__30pm'
2025-06-18 18:13:40.177 [1750263198685669/preprocess_training_data/5 (pid
25836)] 'title_tfidf__3rd' 'title_tfidf__40' 'title_tfidf__50'
'title_tfidf__500']
2025-06-18 18:13:40.177 [1750263198685669/preprocess_training_data/5 (pid
25836)] Processed Training DataFrame created.
2025-06-18 18:13:40.213 [1750263198685669/preprocess_training_data/5 (pid
25836)] title_tfidf__00 ... cat_onehot__location_Zionsville, IN
2025-06-18 18:13:41.755 [1750263198685669/preprocess_training_data/5 (pid
25836)] 15628          0.0 ...          0.0
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] 4015          0.0 ...          0.0
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] 4453          0.0 ...          0.0
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] 8845          0.0 ...          0.0
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] 9195          0.0 ...          0.0
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)]
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid

```

```

25836)] [5 rows x 12895 columns]
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)]
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] Transforming test data...
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] Test data transformed. Shape: (4621, 12895)
2025-06-18 18:13:41.756 [1750263198685669/preprocess_training_data/5 (pid
25836)] Processed Test DataFrame created.
2025-06-18 18:13:41.800 [1750263198685669/preprocess_training_data/5 (pid
25836)] title_tfidf__00 ... cat_onehot__location_Zionsville, IN
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] 22136 0.0 ... 0.0
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] 15278 0.0 ... 0.0
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] 22573 0.0 ... 0.0
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] 8310 0.0 ... 0.0
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] 22669 0.0 ... 0.0
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)]
2025-06-18 18:13:46.317 [1750263198685669/preprocess_training_data/5 (pid
25836)] [5 rows x 12895 columns]
2025-06-18 18:13:46.319 [1750263198685669/preprocess_training_data/5 (pid
25836)] Task finished successfully.
2025-06-18 18:13:46.327 [1750263198685669/train_model/6 (pid 25972)] Task is
starting.
2025-06-18 18:13:47.017 [1750263198685669/train_model/6 (pid 25972)] Attempting
to use MLflow Tracking URI: None
2025-06-18 18:13:48.376 [1750263198685669/train_model/6 (pid 25972)] in mlflow
context
2025-06-18 18:13:48.411 [1750263198685669/train_model/6 (pid 25972)] MLflow Run
ID: e1244438d5ed4c8a9536855272ea6422
2025-06-18 18:13:48.411 [1750263198685669/train_model/6 (pid 25972)] MLflow Run
Name: 1750263198685669
2025-06-18 18:13:48.411 [1750263198685669/train_model/6 (pid 25972)] MLflow
Experiment Dataset Name: linkedin_joblist_salary_less_than_500k_and_above_1k
2025-06-18 18:13:48.411 [1750263198685669/train_model/6 (pid 25972)] Training
Decision Tree Regressor...
2025-06-18 18:13:48.415 [1750263198685669/train_model/6 (pid 25972)] <flow
DTRFlow step train_model> failed:
2025-06-18 18:13:48.425 [1750263198685669/train_model/6 (pid 25972)] Internal
error
2025-06-18 18:13:48.428 [1750263198685669/train_model/6 (pid 25972)] Traceback
(most recent call last):
2025-06-18 18:13:48.428 [1750263198685669/train_model/6 (pid 25972)] File

```



```

/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli.py", line 619, in main
2025-06-18 18:13:48.428 [1750263198685669/train_model/6 (pid 25972)]
start(auto_envvar_prefix="METAFLOW", obj=state)
2025-06-18 18:13:48.428 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 829, in __call__
2025-06-18 18:13:48.428 [1750263198685669/train_model/6 (pid 25972)] return
self.main(args, kwargs)
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 782, in main
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] rv =
self.invoke(ctx)
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli_components/utils.py", line 69, in invoke
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] return
_process_result(sub_ctx.command.invoke(sub_ctx))
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 1066, in invoke
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] return
ctx.invoke(self.callback, ctx.params)
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/_vendor/click/core.py", line 610, in invoke
2025-06-18 18:13:48.625 [1750263198685669/train_model/6 (pid 25972)] return
callback(args, kwargs)
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/tracing/_init_.py", line 27, in wrapper_func
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] return
func(args, kwargs)
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-

```

```

packages/metaflow/_vendor/click/decorators.py", line 21, in new_func
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] return
f(get_current_context(), args, kwargs)
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
~~~~~
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/cli_components/step_cmd.py", line 167, in step
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
task.run_step(
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 672, in run_step
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
self._exec_step_function(step_func)
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 62, in _exec_step_function
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
step_function()
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] File
"/Users/mahdikhashan/tmp/tu-
mlops/pipelines/experiment_1_decision_tree_fault_tolerance.py", line 363, in
train_model
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] raise
Fault("model fitting failed - it should be handled, why!!!")
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
utils.FaultException.Fault: model fitting failed - it should be handled, why!!!
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)]
2025-06-18 18:13:48.626 [1750263198685669/train_model/6 (pid 25972)] Task
failed.
2025-06-18 18:13:48.628 [1750263198685669/train_model/6 (pid 25990)] Task is
starting (retry).
2025-06-18 18:13:49.043 [1750263198685669/train_model/6 (pid 25990)] Attempting
to use MLflow Tracking URI: None
2025-06-18 18:13:49.886 [1750263198685669/train_model/6 (pid 25990)] in mlflow
context
2025-06-18 18:13:49.916 [1750263198685669/train_model/6 (pid 25990)] MLflow Run
ID: a7fcee9fe4a42dfaee0a1f2140c1f25
2025-06-18 18:13:49.917 [1750263198685669/train_model/6 (pid 25990)] MLflow Run
Name: 1750263198685669
2025-06-18 18:13:49.917 [1750263198685669/train_model/6 (pid 25990)] MLflow
Experiment Dataset Name: linkedin_joblist_salary_less_than_500k_and_above_1k
2025-06-18 18:13:49.917 [1750263198685669/train_model/6 (pid 25990)] Training
Decision Tree Regressor...
2025-06-18 18:13:59.527 [1750263198685669/train_model/6 (pid 25990)] Model
training complete.
2025-06-18 18:13:59.907 [1750263198685669/train_model/6 (pid 25990)] Making

```

predictions on the test set...
 2025-06-18 18:13:59.907 [1750263198685669/train_model/6 (pid 25990)]
 2025-06-18 18:13:59.911 [1750263198685669/train_model/6 (pid 25990)] Evaluating
 model performance on the test set:
 2025-06-18 18:13:59.911 [1750263198685669/train_model/6 (pid 25990)] Mean
 Absolute Error (MAE): 36109.40
 2025-06-18 18:14:00.758 [1750263198685669/train_model/6 (pid 25990)] Root Mean
 Squared Error (RMSE): 50543.49
 2025-06-18 18:14:00.758 [1750263198685669/train_model/6 (pid 25990)] R-squared
 (R^2): 0.2377
 2025-06-18 18:14:00.759 [1750263198685669/train_model/6 (pid 25990)]
 2025-06-18 18:14:00.771 [1750263198685669/train_model/6 (pid 25990)] Root Mean
 Squared Error (RMSE) on Training Set: 47574.65
 2025-06-18 18:14:00.771 [1750263198685669/train_model/6 (pid 25990)] logging
 current model version.
 2025-06-18 18:15:44.613 [1750263198685669/train_model/6 (pid 25990)] 2025/06/18
 18:15:44 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use
 `name` instead.
 2025-06-18 18:15:47.462 [1750263198685669/train_model/6 (pid 25990)] Registered
 model 'A_B_second_model' already exists. Creating a new version of this model...
 2025-06-18 18:15:47.467 [1750263198685669/train_model/6 (pid 25990)] Created
 version '2' of model 'A_B_second_model'.
 2025-06-18 18:15:47.472 [1750263198685669/train_model/6 (pid 25990)] Current
 model (R^2 : 0.2377) logged as A_B_second_model.
 2025-06-18 18:15:51.175 [1750263198685669/train_model/6 (pid 25990)] Task
 finished successfully.
 2025-06-18 18:15:51.183 [1750263198685669/evaluate_robustness/7 (pid 26218)]
 Task is starting.
 2025-06-18 18:15:51.613 [1750263198685669/evaluate_robustness/7 (pid 26218)]
 Attempting to use MLflow Tracking URI: None
 2025-06-18 18:15:51.946 [1750263198685669/evaluate_robustness/7 (pid 26218)] R2
 metric found, comparing with the baseline
 2025-06-18 18:15:52.064 [1750263198685669/evaluate_robustness/7 (pid 26218)]
 Task finished successfully.
 2025-06-18 18:15:52.068 [1750263198685669/end/8 (pid 26225)] Task is starting.
 2025-06-18 18:15:52.438 [1750263198685669/end/8 (pid 26225)] Attempting to use
 MLflow Tracking URI: None
 2025-06-18 18:15:52.508 [1750263198685669/end/8 (pid 26225)] Flow 'DTRFlow'
 completed successfully.
 2025-06-18 18:15:52.576 [1750263198685669/end/8 (pid 26225)] Task finished
 successfully.
 2025-06-18 18:15:52.577 Done!
 Run('DTRFlow/1750263198685669') finished

0.1.3 List artifacts of experiments

```
[ ]: # TODO(mahdi): ...
```

0.1.4 run A/B test

```
[55]: a_b_test_flow = "pipelines/evaluate_A_B.py"
      model_a_uri="models:/A_B_second_model/2"
      model_b_uri="models:/A_B_control_model/1"
      dataset_a_b = "data/A_B_1.parquet"
      with Runner(a_b_test_flow).run(dataset=dataset_a_b, model_a_uri=model_a_uri,
      ↪model_b_uri=model_b_uri) as running:
          print(f'{running.run} finished')
```

Metaflow 2.15.9 executing Evaluate_A_B_Flow for user:mahdikhashan

Validating your flow...

The graph looks good!

Running pylint...

Pylint not found, so extra checks are disabled.

Including file data/A_B_1.parquet of size 6MB

2025-06-18 20:42:43.173 Workflow starting (run-id 1750272162974094):

2025-06-18 20:42:43.182 [1750272162974094/start/1 (pid 48502)] Task is starting.

2025-06-18 20:42:43.377 [1750272162974094/start/1 (pid 48502)] Task finished

successfully.

2025-06-18 20:42:43.382 [1750272162974094/load_dataset/2 (pid 48509)] Task is starting.

2025-06-18 20:42:44.284 [1750272162974094/load_dataset/2 (pid 48509)] Task finished successfully.

2025-06-18 20:42:44.290 [1750272162974094/validate_dataset/3 (pid 48516)] Task is starting.

2025-06-18 20:42:44.822 [1750272162974094/validate_dataset/3 (pid 48516)]

===== test session starts =====

2025-06-18 20:42:44.823 [1750272162974094/validate_dataset/3 (pid 48516)]

platform darwin -- Python 3.11.10, pytest-8.3.2, pluggy-1.6.0 --

/Users/mahdikhashan/tmp/tu-mlops/.venv/bin/python

2025-06-18 20:42:44.823 [1750272162974094/validate_dataset/3 (pid 48516)]

cachedir: .pytest_cache

2025-06-18 20:42:46.155 [1750272162974094/validate_dataset/3 (pid 48516)]

rootdir: /Users/mahdikhashan/tmp/tu-mlops

2025-06-18 20:42:46.155 [1750272162974094/validate_dataset/3 (pid 48516)]

configfile: pyproject.toml

2025-06-18 20:42:46.155 [1750272162974094/validate_dataset/3 (pid 48516)]

plugins: anyio-4.9.0

2025-06-18 20:42:46.155 [1750272162974094/validate_dataset/3 (pid 48516)]

collecting ... 1.3.11

2025-06-18 20:42:46.164 [1750272162974094/validate_dataset/3 (pid 48516)]

collected 5 items

2025-06-18 20:42:46.165 [1750272162974094/validate_dataset/3 (pid 48516)]

```

2025-06-18 20:42:46.250 [1750272162974094/validate_dataset/3 (pid 48516)]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/8 [00:00<?, ?it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 1583.35it/s]
Calculating Metrics: 25%|         | 2/8 [00:00<00:00, 1196.83it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 1714.29it/s]
Calculating Metrics: 38%|         | 3/8 [00:00<00:00, 1673.26it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 1828.98it/s]
Calculating Metrics: 50%|         | 4/8 [00:00<00:00, 1794.55it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2145.84it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2114.60it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2099.78it/s]
Calculating Metrics: 100%|        | 8/8 [00:00<00:00, 2076.90it/s]
2025-06-18 20:42:46.254 [1750272162974094/validate_dataset/3 (pid 48516)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_title PASSED
2025-06-18 20:42:46.309 [1750272162974094/validate_dataset/3 (pid 48516)]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 15621.24it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 10922.67it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 14217.98it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 12087.33it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 9341.43it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 8605.47it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11991.63it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11638.41it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11423.83it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11111.87it/s]
2025-06-18 20:42:46.313 [1750272162974094/validate_dataset/3 (pid 48516)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_description
PASSED
2025-06-18 20:42:46.365 [1750272162974094/validate_dataset/3 (pid 48516)]
tests/test_validate_dataset.py::test_validate_max_salary_in_range PASSED
2025-06-18 20:42:46.416 [1750272162974094/validate_dataset/3 (pid 48516)]
tests/test_validate_dataset.py::test_validate_min_salary_in_range PASSED
2025-06-18 20:42:46.416 [1750272162974094/validate_dataset/3 (pid 48516)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 19065.02it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 14388.69it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 18183.40it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 16215.09it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 14388.69it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 13336.42it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 12623.94it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 11658.94it/s]
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
tests/test_validate_dataset.py::test_validate_column_values_in_list PASSED

```

```

2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
Calculating Metrics: 0%|          | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 40%|         | 2/5 [00:00<00:00, 18682.87it/s]
Calculating Metrics: 40%|         | 2/5 [00:00<00:00, 13530.01it/s]
Calculating Metrics: 60%|         | 3/5 [00:00<00:00, 17142.93it/s]
Calculating Metrics: 60%|         | 3/5 [00:00<00:00, 15215.13it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 16181.73it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 14926.35it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 14122.24it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 13058.23it/s]
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
===== warnings summary =====
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 18850.80it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 13981.01it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 17747.41it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 15787.84it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 1565.04it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 1547.29it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 1535.39it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 1518.03it/s]
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095:
ChangedInMarshmallow4Warning: `Number` field should not be instantiated. Use
`Integer`, `Float`, or `Decimal` instead.
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
config_version: fields.Number = fields.Number(
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)]
2025-06-18 20:42:46.471 [1750272162974094/validate_dataset/3 (pid 48516)] --
Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
2025-06-18 20:42:47.284 [1750272162974094/validate_dataset/3 (pid 48516)]
===== 5 passed, 1 warning in 1.65s =====
2025-06-18 20:42:47.285 [1750272162974094/validate_dataset/3 (pid 48516)]
Foreach yields 2 child steps.
2025-06-18 20:42:47.285 [1750272162974094/validate_dataset/3 (pid 48516)] Task
finished successfully.
2025-06-18 20:42:47.290 [1750272162974094/predict/4 (pid 48524)] Task is
starting.
2025-06-18 20:42:47.294 [1750272162974094/predict/5 (pid 48525)] Task is
starting.

```



```
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====
=====+=====+=====+=====+
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] | A_B_second_model
|      2 | company_name: Lakefield Veterinary Group
|    32240 |      50167.6 | 17927.6 |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          | title: Customer Service Rep
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          | description: CUSTOMER SERVICE REPRESENTATIVE (CSR)
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          |
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          | About us: Veterinary Medical Center of The Woodlands is a full-
service veterinary hospital providing excellent medical, surgical, and dental
care. We have an in-house pharmacy, laboratory, and radiology equipment.
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          |
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          | Our goal has been to assemble a veterinary health care team
committed to providing quality care, exceptional client service, and a team that
believes in treating your pets like the valued family members they are.
|          |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|          |
|          |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|          | Find out more about us here: VMC of the Woodlands
```

```

=====+=====+=====+
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] | A_B_second_model
|           2 | company_name: Lakefield Veterinary Group
|    32240 |    50167.6 | 17927.6 |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           | title: Customer Service Rep
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           | description: CUSTOMER SERVICE REPRESENTATIVE (CSR)
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           |
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           | About us: Veterinary Medical Center of The Woodlands is a full-
service veterinary hospital providing excellent medical, surgical, and dental
care. We have an in-house pharmacy, laboratory, and radiology equipment.
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           |
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           | Our goal has been to assemble a veterinary health care team
committed to providing quality care, exceptional client service, and a team that
believes in treating your pets like the valued family members they are.
|           |           |           |
2025-06-18 20:42:51.406 [1750272162974094/join/6 (pid 48555)] |
|           |
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           | Find out more about us here: VMC of the Woodlands

```



```

|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           | POSITION SUMMARY: The Customer Service Representative (CSR)
actively recruits new clients and promotes hospital services. Schedules client
visits and drives the flow of clients and pets through the hospital to aid the
productivity of the veterinary medical team. Ensures good communication with the
clients, and coordinates the care of the pets, influencing clients to return and
refer their friends and families.
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           | WHAT YOU'LL DO (Essential Functions):
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           | Foster a culture built on our values: Respect, Learning, Do the
Right Thing, WOW Service, and FunMaintain knowledge of the services the hospital
provides and utilizes the information to promote the hospital's services and
products to existing and prospective customersWelcome clients and pets, ensuring
that they are comfortable in the hospital, and educating the clients about
preventative care and pet health needsAssist incoming clients. Document and
enter the pet's history, requested services, and other pertinent information
within the practice management softwareDirect the flow of clients and pets to
ensure superior client care and maximum productivity of the veterinary medical
teamAnswer basic questions from clients on the phone and in person, act as an
initial triage to determine patient's service needs. Refer the patient to the
appropriate veterinary staff as neededSchedule client appointments and follow-up
exams. Work with Veterinarian(s) and veterinary staff as needed to ensure
optimal schedulingAssist outgoing clients by providing all necessary
instructions, information, and invoicesDispense prescription items per the
Veterinarian's instructions and sells retail productsResponsible for cash
management including accepting payments for products and services provided.
Perform daily reconciliation of the cash drawerRespond to customer complaints in
a professional, courteous manner and act satisfactorily to resolve the
problemSupport a practice environment that promotes collaboration and teamwork
to ensure the delivery of quality, compassionate, effective, and efficient
health care to petsComply with all practice policies, quality assurance
guidelines, and minimum standards of carePerform other duties and projects as
requested |           |           |
2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
|           |           |           |

```

2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		WHAT WE'RE LOOKING FOR (Essential Qualifications):	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		Education:	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		High School Diploma or GED required Professional Licenses/Certifications: None	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		Experience:	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		1+ years experience in retail sales or customer service preferredExperience in a medical facility desired	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	
		Knowledge, Skills and Ability:	
2025-06-18	20:42:51.407	[1750272162974094/join/6 (pid 48555)]	

2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
 | Ability to provide effective customer service principlesBasic
 knowledge of medical terminologyBasic proficiency with a variety of computer
 programs including Microsoft Office SuiteBasic written, verbal, and active
 listening communication skillsStrong customer relations and interpersonal
 skillsExercises tact and diplomacy in dealing with sensitive issues and
 situationsPrioritize work and handle a variety of tasks simultaneously, with
 frequent interruptionsStrong organizational and time management
 skillsEffectively and quickly responds to requests in a proactive mannerStrong
 attention to detail and high level of accuracy
 | | | |
 2025-06-18 20:42:51.407 [1750272162974094/join/6 (pid 48555)] |
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | PHYSICAL DEMANDS:
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | While performing the duties of this job, the team member is
 regularly required to talk and hear. The team member must frequently use a
 computer for long periods of time. Specific vision abilities are required to do
 this job including close vision. This position requires walking and/or standing
 for long periods of time, bending, reaching, grasping, and repetitive motions.
 May be exposed to disinfectant/sanitation chemicals, pets dander, and
 excretions. This position is exposed to low to moderate noise levels. Must be
 able to lift up to 30 lbs. unassisted, 50 lbs. with assistance. Must be able to
 work flexible hours including evenings and weekends.
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | | | |
 2025-06-18 20:42:51.408 [1750272162974094/join/6 (pid 48555)] |
 | SUPERVISORY RESPONSIBILITIES: None
 | | | |
 2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |
 | location: The Woodlands, TX
 | | | |
 2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |
 | remote_allowed: 0
 | | | |
 2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |
 | work_type: FULL_TIME

[illegible]

believes in treating your pets like the valued family members they are.

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | Find out more about us here: VMC of the Woodlands

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | POSITION SUMMARY: The Customer Service Representative (CSR) actively recruits new clients and promotes hospital services. Schedules client visits and drives the flow of clients and pets through the hospital to aid the productivity of the veterinary medical team. Ensures good communication with the clients, and coordinates the care of the pets, influencing clients to return and refer their friends and families.

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | WHAT YOU'LL DO (Essential Functions):

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| | Foster a culture built on our values: Respect, Learning, Do the Right Thing, WOW Service, and FunMaintain knowledge of the services the hospital provides and utilizes the information to promote the hospital's services and products to existing and prospective customersWelcome clients and pets, ensuring that they are comfortable in the hospital, and educating the clients about preventative care and pet health needsAssist incoming clients. Document and enter the pet's history, requested services, and other pertinent information within the practice management softwareDirect the flow of clients and pets to ensure superior client care and maximum productivity of the veterinary medical teamAnswer basic questions from clients on the phone and in person, act as an initial triage to determine patient's service needs. Refer the patient to the appropriate veterinary staff as neededSchedule client appointments and follow-up exams. Work with Veterinarian(s) and veterinary staff as needed to ensure optimal schedulingAssist outgoing clients by providing all necessary instructions, information, and invoicesDispense prescription items per the Veterinarian's instructions and sells retail productsResponsible for cash management including accepting payments for products and services provided. Perform daily reconciliation of the cash drawerRespond to customer complaints in a professional, courteous manner and act satisfactorily to resolve the

problemSupport a practice environment that promotes collaboration and teamwork
to ensure the delivery of quality, compassionate, effective, and efficient
health care to petsComply with all practice policies, quality assurance
guidelines, and minimum standards of carePerform other duties and projects as
requested | | |

2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| |
| | | |
2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |

| |
| | | |
2025-06-18 20:42:51.409 [1750272162974094/join/6 (pid 48555)] |
| | WHAT WE'RE LOOKING FOR (Essential Qualifications):

| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |

| |
| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |
| | Education:

| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |
| | | |

2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |
| | High School Diploma or GED required Professional
Licenses/Certifications: None

| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |
| | | |

2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |
| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |

| | Experience:
| | | |
2025-06-18 20:42:51.410 [1750272162974094/join/6 (pid 48555)] |

| | | |
2025-06-18 20:42:51.411 [1750272162974094/join/6 (pid 48555)] |
| | 1+ years experience in retail sales or customer service
preferredExperience in a medical facility desired

| | | |
2025-06-18 20:42:51.411 [1750272162974094/join/6 (pid 48555)] |
| | | |

2025-06-18 20:42:51.411 [1750272162974094/join/6 (pid 48555)] |
| |

2025-06-18	20:42:51.411	[1750272162974094/join/6 (pid 48555)]	
			Knowledge, Skills and Ability:
2025-06-18	20:42:51.411	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.411	[1750272162974094/join/6 (pid 48555)]	
			Ability to provide effective customer service principlesBasic knowledge of medical terminologyBasic proficiency with a variety of computer programs including Microsoft Office SuiteBasic written, verbal, and active listening communication skillsStrong customer relations and interpersonal skillsExercises tact and diplomacy in dealing with sensitive issues and situationsPrioritize work and handle a variety of tasks simultaneously, with frequent interruptionsStrong organizational and time management skillsEffectively and quickly responds to requests in a proactive mannerStrong attention to detail and high level of accuracy
2025-06-18	20:42:51.411	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.412	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.412	[1750272162974094/join/6 (pid 48555)]	
			PHYSICAL DEMANDS:
2025-06-18	20:42:51.456	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.456	[1750272162974094/join/6 (pid 48555)]	
			While performing the duties of this job, the team member is regularly required to talk and hear. The team member must frequently use a computer for long periods of time. Specific vision abilities are required to do this job including close vision. This position requires walking and/or standing for long periods of time, bending, reaching, grasping, and repetitive motions. May be exposed to disinfectant/sanitation chemicals, pets dander, and excretions. This position is exposed to low to moderate noise levels. Must be able to lift up to 30 lbs. unassisted, 50 lbs. with assistance. Must be able to work flexible hours including evenings and weekends.
2025-06-18	20:42:51.456	[1750272162974094/join/6 (pid 48555)]	
2025-06-18	20:42:51.456	[1750272162974094/join/6 (pid 48555)]	
			SUPERVISORY RESPONSIBILITIES: None
2025-06-18	20:42:51.456	[1750272162974094/join/6 (pid 48555)]	


```
print(f'{running.run} finished')
```

Metaflow 2.15.9 executing EvaluateExperimentFlow for user:mahdikhashan

Validating your flow...

The graph looks good!

Running pylint...

Pylint not found, so extra checks are disabled.

Including file data/A_B_1.parquet of size 6MB

2025-06-18 20:43:36.917 Workflow starting (run-id 1750272216734687):

2025-06-18 20:43:36.925 [1750272216734687/start/1 (pid 48612)] Task is starting.

2025-06-18 20:43:37.905 [1750272216734687/start/1 (pid 48612)] Task finished successfully.

2025-06-18 20:43:37.911 [1750272216734687/load_dataset/2 (pid 48619)] Task is starting.

2025-06-18 20:43:38.449 [1750272216734687/load_dataset/2 (pid 48619)] Task finished successfully.

2025-06-18 20:43:38.453 [1750272216734687/validate_dataset/3 (pid 48626)] Task is starting.

2025-06-18 20:43:38.962 [1750272216734687/validate_dataset/3 (pid 48626)]

===== test session starts =====

2025-06-18 20:43:38.962 [1750272216734687/validate_dataset/3 (pid 48626)]

platform darwin -- Python 3.11.10, pytest-8.3.2, pluggy-1.6.0 --

/Users/mahdikhashan/tmp/tu-mlops/.venv/bin/python

2025-06-18 20:43:38.963 [1750272216734687/validate_dataset/3 (pid 48626)]

cachedir: .pytest_cache

2025-06-18 20:43:40.076 [1750272216734687/validate_dataset/3 (pid 48626)]

rootdir: /Users/mahdikhashan/tmp/tu-mlops

2025-06-18 20:43:40.076 [1750272216734687/validate_dataset/3 (pid 48626)]

configfile: pyproject.toml

2025-06-18 20:43:40.076 [1750272216734687/validate_dataset/3 (pid 48626)]

plugins: anyio-4.9.0

2025-06-18 20:43:40.076 [1750272216734687/validate_dataset/3 (pid 48626)]

collecting ... 1.3.11

2025-06-18 20:43:40.083 [1750272216734687/validate_dataset/3 (pid 48626)]

collected 5 items

2025-06-18 20:43:40.084 [1750272216734687/validate_dataset/3 (pid 48626)]

2025-06-18 20:43:40.161 [1750272216734687/validate_dataset/3 (pid 48626)]

Calculating Metrics: 0%| | 0/8 [00:00<?, ?it/s]

Calculating Metrics: 0%| | 0/8 [00:00<?, ?it/s]

Calculating Metrics: 25%| | 2/8 [00:00<00:00, 1694.67it/s]

Calculating Metrics: 25%| | 2/8 [00:00<00:00, 1527.98it/s]

Calculating Metrics: 38%| | 3/8 [00:00<00:00, 2223.52it/s]

Calculating Metrics: 38%| | 3/8 [00:00<00:00, 2168.72it/s]

Calculating Metrics: 50%| | 4/8 [00:00<00:00, 2190.52it/s]

Calculating Metrics: 50%| | 4/8 [00:00<00:00, 2136.41it/s]

Calculating Metrics: 100%| | 8/8 [00:00<00:00, 3173.30it/s]

Calculating Metrics: 100%| | 8/8 [00:00<00:00, 3129.49it/s]

Calculating Metrics: 100%| | 8/8 [00:00<00:00, 3102.01it/s]

```

Calculating Metrics: 100%|          | 8/8 [00:00<00:00, 3032.48it/s]
2025-06-18 20:43:40.165 [1750272216734687/validate_dataset/3 (pid 48626)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_title PASSED
2025-06-18 20:43:40.228 [1750272216734687/validate_dataset/3 (pid 48626)]
tests/test_validate_dataset.py::test_validate_no_missing_values_in_description
PASSED
2025-06-18 20:43:40.228 [1750272216734687/validate_dataset/3 (pid 48626)]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/13 [00:00<?, ?it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 17084.74it/s]
Calculating Metrics: 15%|         | 2/13 [00:00<00:00, 11983.73it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 15458.12it/s]
Calculating Metrics: 23%|         | 3/13 [00:00<00:00, 13107.20it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 8531.94it/s]
Calculating Metrics: 38%|         | 5/13 [00:00<00:00, 7949.78it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11414.27it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 11102.82it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 10916.11it/s]
Calculating Metrics: 100%|        | 13/13 [00:00<00:00, 10639.21it/s]
2025-06-18 20:43:40.277 [1750272216734687/validate_dataset/3 (pid 48626)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 19065.02it/s]
Calculating Metrics: 50%|         | 2/4 [00:00<00:00, 14169.95it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 17848.10it/s]
Calculating Metrics: 75%|         | 3/4 [00:00<00:00, 15867.48it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 14327.26it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 13252.15it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 12539.03it/s]
Calculating Metrics: 100%|        | 4/4 [00:00<00:00, 11562.52it/s]
2025-06-18 20:43:40.278 [1750272216734687/validate_dataset/3 (pid 48626)]
tests/test_validate_dataset.py::test_validate_max_salary_in_range PASSED
2025-06-18 20:43:40.324 [1750272216734687/validate_dataset/3 (pid 48626)]
Calculating Metrics: 0%|          | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/5 [00:00<?, ?it/s]
Calculating Metrics: 40%|         | 2/5 [00:00<00:00, 18850.80it/s]
Calculating Metrics: 40%|         | 2/5 [00:00<00:00, 14098.50it/s]
Calculating Metrics: 60%|         | 3/5 [00:00<00:00, 17873.45it/s]
Calculating Metrics: 60%|         | 3/5 [00:00<00:00, 15709.00it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 16777.22it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 15522.96it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 14665.40it/s]
Calculating Metrics: 100%|        | 5/5 [00:00<00:00, 13512.58it/s]
2025-06-18 20:43:40.326 [1750272216734687/validate_dataset/3 (pid 48626)]
tests/test_validate_dataset.py::test_validate_min_salary_in_range PASSED
2025-06-18 20:43:40.374 [1750272216734687/validate_dataset/3 (pid 48626)]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]
Calculating Metrics: 0%|          | 0/4 [00:00<?, ?it/s]

```

```

Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 19737.90it/s]
Calculating Metrics: 50%|          | 2/4 [00:00<00:00, 14665.40it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 18613.78it/s]
Calculating Metrics: 75%|          | 3/4 [00:00<00:00, 16469.78it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 5855.92it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 5639.40it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 5493.52it/s]
Calculating Metrics: 100%|         | 4/4 [00:00<00:00, 5284.16it/s]
2025-06-18 20:43:40.376 [1750272216734687/validate_dataset/3 (pid 48626)]
tests/test_validate_dataset.py::test_validate_column_values_in_list PASSED
2025-06-18 20:43:40.376 [1750272216734687/validate_dataset/3 (pid 48626)]
2025-06-18 20:43:40.376 [1750272216734687/validate_dataset/3 (pid 48626)]
===== warnings summary =====
2025-06-18 20:43:40.376 [1750272216734687/validate_dataset/3 (pid 48626)]
.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095
2025-06-18 20:43:40.376 [1750272216734687/validate_dataset/3 (pid 48626)]
/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/great_expectations/data_context/types/base.py:1095:
ChangedInMarshmallow4Warning: `Number` field should not be instantiated. Use
`Integer`, `Float`, or `Decimal` instead.
2025-06-18 20:43:40.377 [1750272216734687/validate_dataset/3 (pid 48626)]
config_version: fields.Number = fields.Number(
2025-06-18 20:43:40.377 [1750272216734687/validate_dataset/3 (pid 48626)]
2025-06-18 20:43:40.377 [1750272216734687/validate_dataset/3 (pid 48626)] --
Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
2025-06-18 20:43:40.377 [1750272216734687/validate_dataset/3 (pid 48626)]
===== 5 passed, 1 warning in 1.41s =====
2025-06-18 20:43:41.153 [1750272216734687/validate_dataset/3 (pid 48626)]
Foreach yields 3 child steps.
2025-06-18 20:43:41.153 [1750272216734687/validate_dataset/3 (pid 48626)] Task
finished successfully.
2025-06-18 20:43:41.157 [1750272216734687/predict/4 (pid 48634)] Task is
starting.
2025-06-18 20:43:41.160 [1750272216734687/predict/5 (pid 48635)] Task is
starting.
2025-06-18 20:43:41.163 [1750272216734687/predict/6 (pid 48636)] Task is
starting.
2025-06-18 20:43:44.451 [1750272216734687/predict/5 (pid 48635)] Downloading
artifacts: 0%|          | 0/1 [00:00<?, ?it/s]
Downloading artifacts: 100%|        | 1/1 [00:00<00:00, 3331.46it/s]
Downloading artifacts: 100%|        | 1/1 [00:00<00:00, 2427.26it/s]
2025-06-18 20:43:44.456 [1750272216734687/predict/6 (pid 48636)] Downloading
artifacts: 0%|          | 0/1 [00:00<?, ?it/s]
Downloading artifacts: 0%|          | 0/1 [00:00<?, ?it/s]
2025-06-18 20:43:44.457 [1750272216734687/predict/4 (pid 48634)] Downloading
artifacts: 0%|          | 0/1 [00:00<?, ?it/s]
Downloading artifacts: 100%|        | 1/1 [00:00<00:00, 5178.15it/s]

```


2025-06-18 20:43:45.459 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.459 [1750272216734687/join/7 (pid 48678)] |
| | Successfully learn and sell all Aftermarket products and offerings, to include but not limited to, Parts, Service, Contract Maintenance, technology innovation products, and Rush's RushCare offerings (Service Connect and PartsConnect).Competence in systematic selling, which includes pre-call planning, investigating sales opportunities, offering Rush specific solutions, and closing business sales.Call on National Accounts through various mediums, partnering closely with Rush Aftermarket National Accounts Sales Manager (RSM) to promote account programs and process.Sell multiple Parts product categories and specific Service repair types.Promote and sell "All Makes" Parts and Service, beyond proprietary brands.Promote and sell Rush and OE special market programs.Develop, implement, and maintain comprehensive account specific growth strategies for all products offered.Participate in company, market, or vendor driven sales blitzes, trade shows and annual job specific employee and customer events, to include periodic training and development off site. |
|
2025-06-18 20:43:45.459 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.459 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.460 [1750272216734687/join/7 (pid 48678)] |
| | Basic Qualifications:
|
2025-06-18 20:43:45.462 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.462 [1750272216734687/join/7 (pid 48678)] |
| | High school diploma or general education degree (GED). Prefer Bachelors' degree from an accredited university or college. Experienced in business to business sales, or within our industry preferred.Must possess a current and valid driver license with a driving record that meets the insurability guidelines set forth by the company's insurance carrier.Gain literacy and competency of Microsoft Dynamics 360 (CRM) to pre-call plan, document sales strategy, manage all accounts sales activity.Capable of managing an assigned Book of Business (BOB), which will include 40-50 accounts.
|
|
2025-06-18 20:43:45.462 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.462 [1750272216734687/join/7 (pid 48678)] |
|
|
2025-06-18 20:43:45.462 [1750272216734687/join/7 (pid 48678)] |

		Benefits:	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		We offer exceptional compensation and benefits, 401K and stock	
		purchase, incentives for performance, training, and opportunity for advancement	
		- all in a culture that appreciates and rewards excellence, a positive attitude	
		and integrity.	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		Equal Opportunity Employer Minorities/Women/Protected	
		Veterans/Disabled	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		This job posting does not state or imply that these are the only	
		duties to be performed by this employee. The employee will be required to follow	
		any other instructions and to perform any other duties requested by their	
		leader.	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		Minimum Pay Rate	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		USD \$60,000.00/Hr.	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
2025-06-18	20:43:45.462	[1750272216734687/join/7 (pid 48678)]	
		Maximum Pay Rate	

[illegible]

representing truck and bus manufacturers. Our franchises include Peterbilt, International, Hino, Isuzu, Ford, IC Bus and Blue Bird. We offer a rewarding career as a leader in the transportation industry. Grow with us as we continue to expand our network of locations and services.

|
2025-06-18 20:43:45.463 [1750272216734687/join/7 (pid 48678)] |

|
|
2025-06-18 20:43:45.463 [1750272216734687/join/7 (pid 48678)] |

| Responsibilities:

|
2025-06-18 20:43:45.463 [1750272216734687/join/7 (pid 48678)] |

|
|
2025-06-18 20:43:45.463 [1750272216734687/join/7 (pid 48678)] |
| Successfully learn and sell all Aftermarket products and offerings, to include but not limited to, Parts, Service, Contract Maintenance, technology innovation products, and Rush's RushCare offerings (Service Connect and PartsConnect).Competence in systematic selling, which includes pre-call planning, investigating sales opportunities, offering Rush specific solutions, and closing business sales.Call on National Accounts through various mediums, partnering closely with Rush Aftermarket National Accounts Sales Manager (RSM) to promote account programs and process.Sell multiple Parts product categories and specific Service repair types.Promote and sell "All Makes" Parts and Service, beyond proprietary brands.Promote and sell Rush and OE special market programs.Develop, implement, and maintain comprehensive account specific growth strategies for all products offered.Participate in company, market, or vendor driven sales blitzes, trade shows and annual job specific employee and customer events, to include periodic training and development off site. |

|
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

|
|
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

|
|
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| Basic Qualifications:

|
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

|
|
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| High school diploma or general education degree (GED). Prefer Bachelors' degree from an accredited university or college. Experienced in business to business sales, or within our industry preferred.Must possess a current and valid driver license with a driving record that meets the insurability guidelines set forth by the company's insurance carrier.Gain

literacy and competency of Microsoft Dynamics 360 (CRM) to pre-call plan, document sales strategy, manage all accounts sales activity. Capable of managing an assigned Book of Business (BOB), which will include 40-50 accounts.

| | | |
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| | Benefits: | |
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.464 [1750272216734687/join/7 (pid 48678)] |
| | We offer exceptional compensation and benefits, 401K and stock purchase, incentives for performance, training, and opportunity for advancement - all in a culture that appreciates and rewards excellence, a positive attitude and integrity.

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |
| | Equal Opportunity Employer Minorities/Women/Protected Veterans/Disabled

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |
| | This job posting does not state or imply that these are the only duties to be performed by this employee. The employee will be required to follow any other instructions and to perform any other duties requested by their leader.

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |
| | Minimum Pay Rate

| | | |
2025-06-18 20:43:45.465 [1750272216734687/join/7 (pid 48678)] |


```

2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 672, in run_step
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)]
self._exec_step_function(step_func)
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] File
"/Users/mahdikhashan/tmp/tu-mlops/.venv/lib/python3.11/site-
packages/metaflow/task.py", line 62, in _exec_step_function
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] step_function()
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] File
"/Users/mahdikhashan/tmp/tu-mlops/pipelines/evaluate_experiment.py", line 106,
in predict
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] raise
FileNotFoundError(
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)] FileNotFoundError:
Artifact 'preprocessor.joblib' not found for run
e6e77a6b75084816955be6952d9ec708 (associated with dtr-tiny v1). Ensure the
preprocessor was logged with this name during model training.
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)]
2025-06-18 20:43:45.519 [1750272216734687/join/7 (pid 48678)]
-----
2025-06-18 20:43:45.520 [1750272216734687/join/7 (pid 48678)] Task finished
successfully.
2025-06-18 20:43:45.524 [1750272216734687/end/8 (pid 48685)] Task is starting.
2025-06-18 20:43:45.702 [1750272216734687/end/8 (pid 48685)] Task finished
successfully.
2025-06-18 20:43:45.703 Done!
Run('EvaluateExperimentFlow/1750272216734687') finished

```

[]: